

Assigment 1 -Human Resource

Yann OBROU

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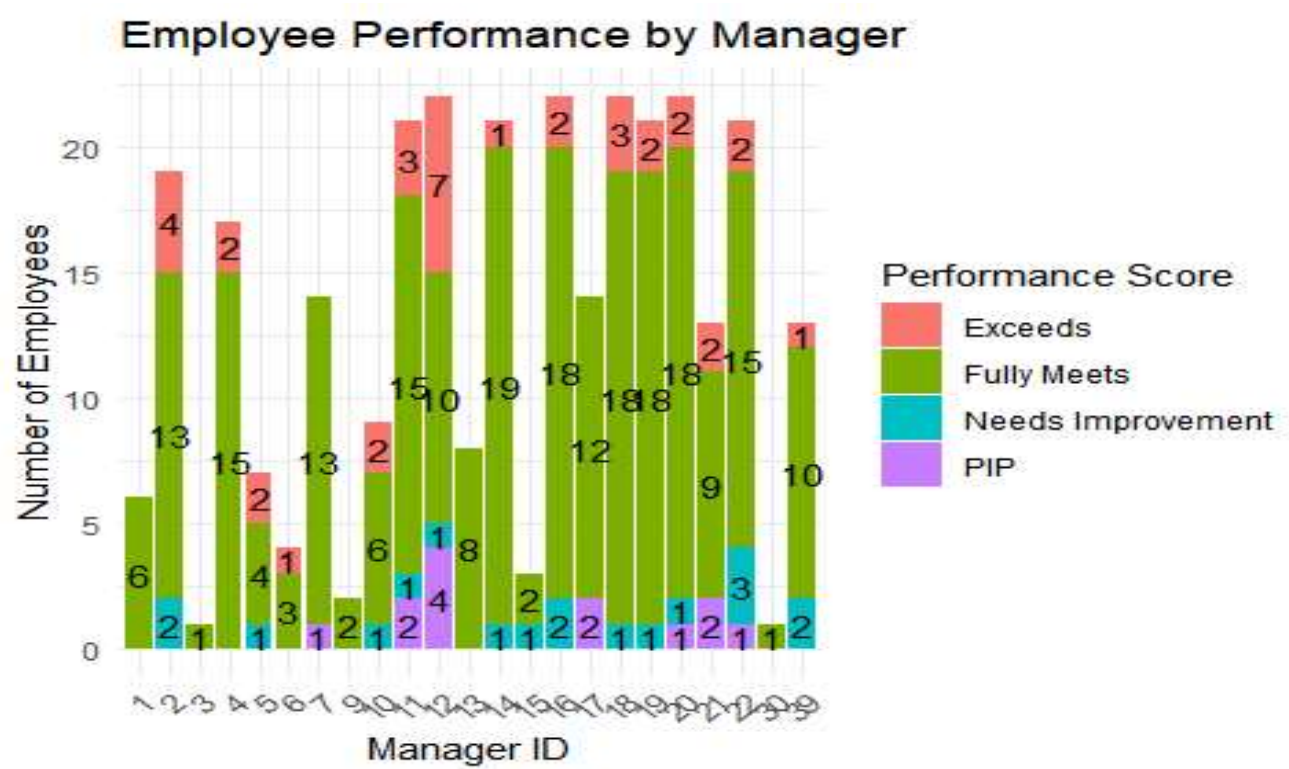
STAT4601/STAT5703 Assignment #1 Due Date: January 30, 2025 midnight

Comprehensive HR Analytics Report: Insights from Visual Data

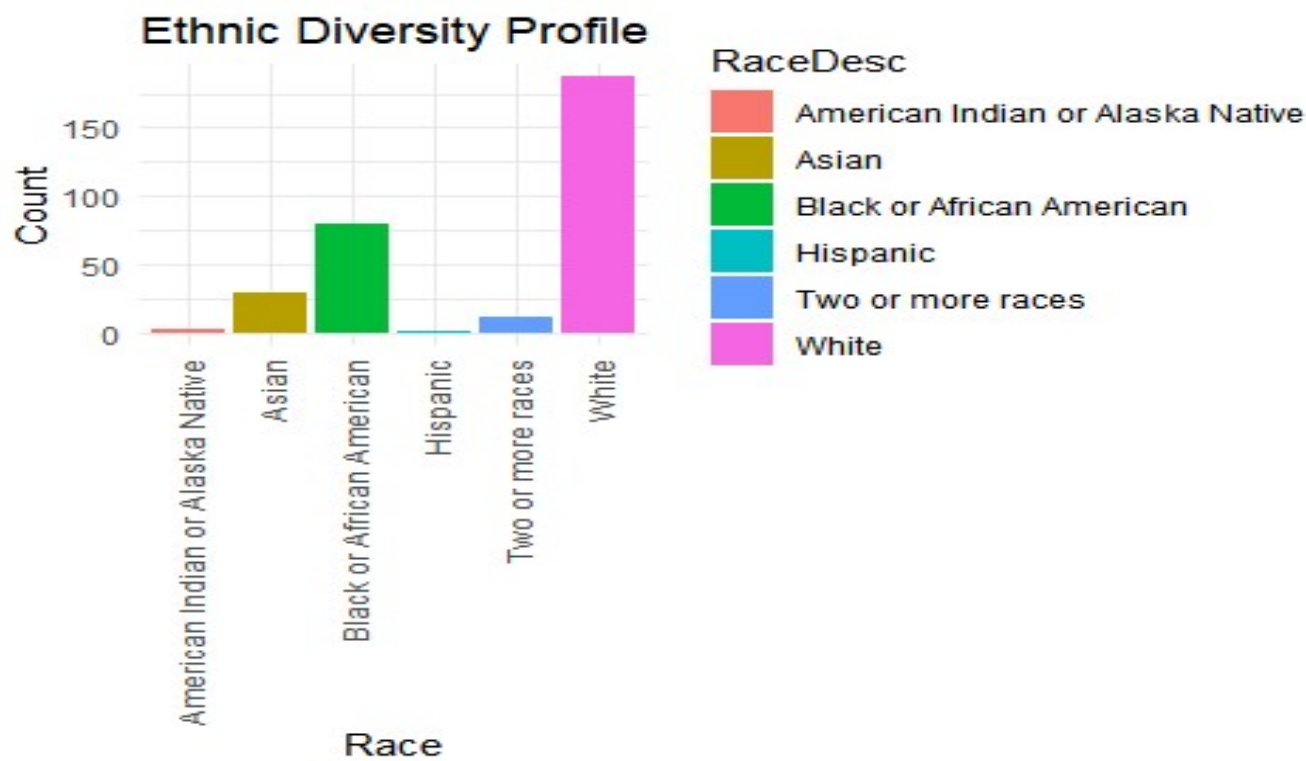
Introduction

Utilizing a robust dataset from the Human Resources department, this report leverages visual analytics to provide comprehensive insights into employee performance, diversity, recruitment efficacy, termination patterns, and pay equity within the organization. Through a detailed exploration of visualizations created in R, this analysis seeks to answer critical questions affecting strategic HR decisions.

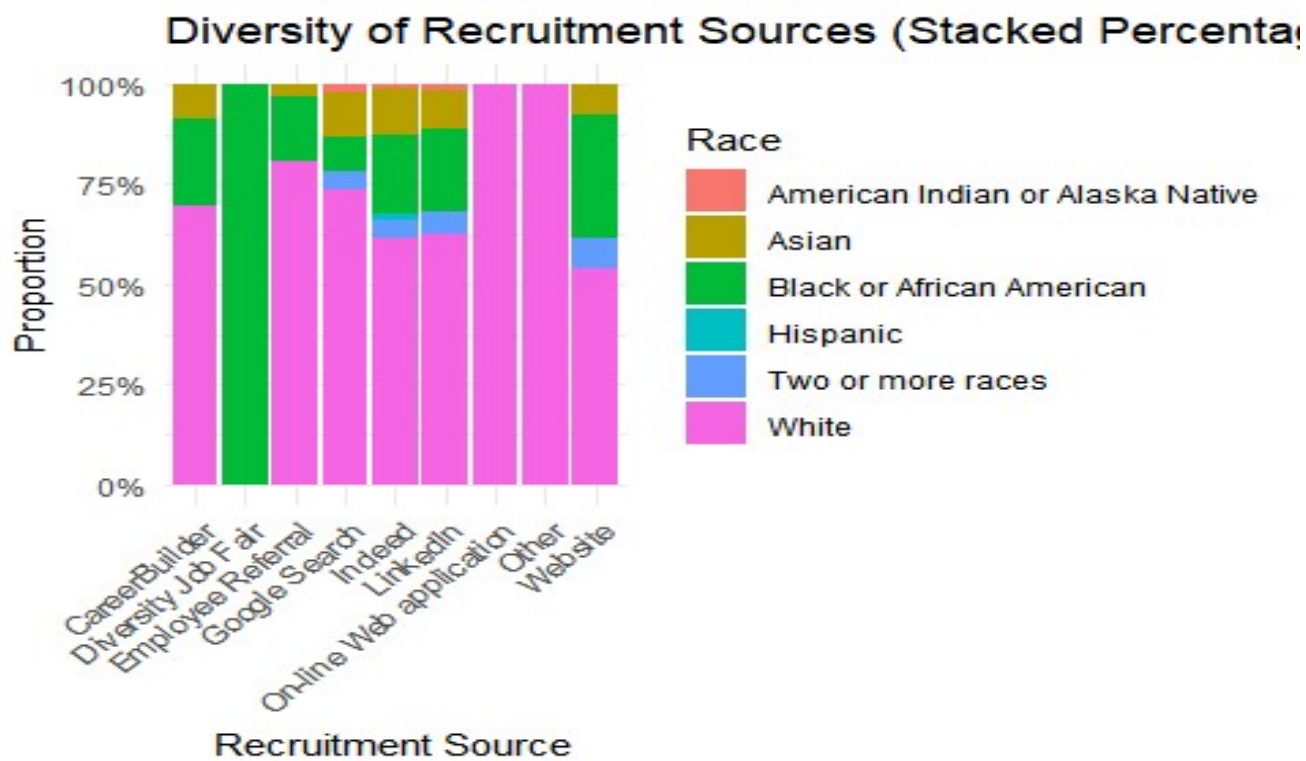
- a) Is there any relationship between who a person works for and their performance score?



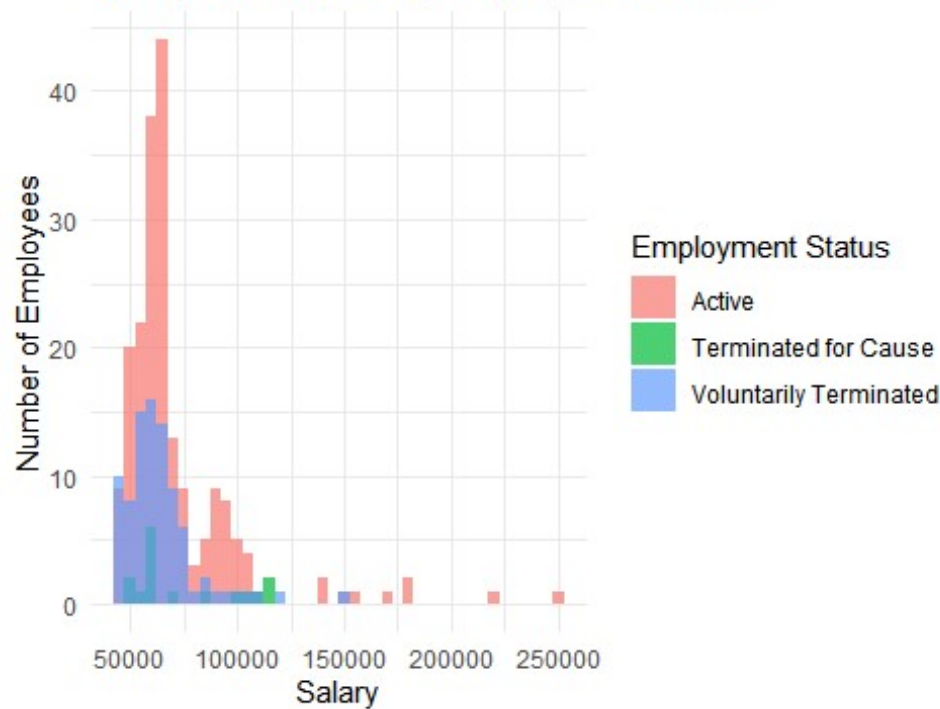
b) What is the overall diversity profile of the organization?



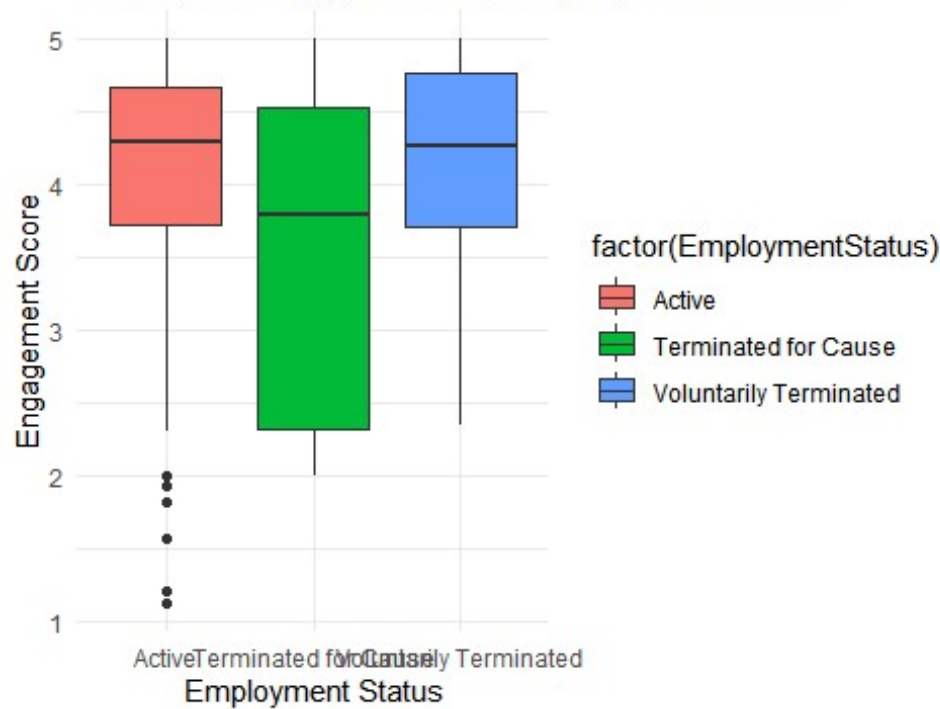
c) What are our best recruiting sources if we want to ensure a diverse organization?

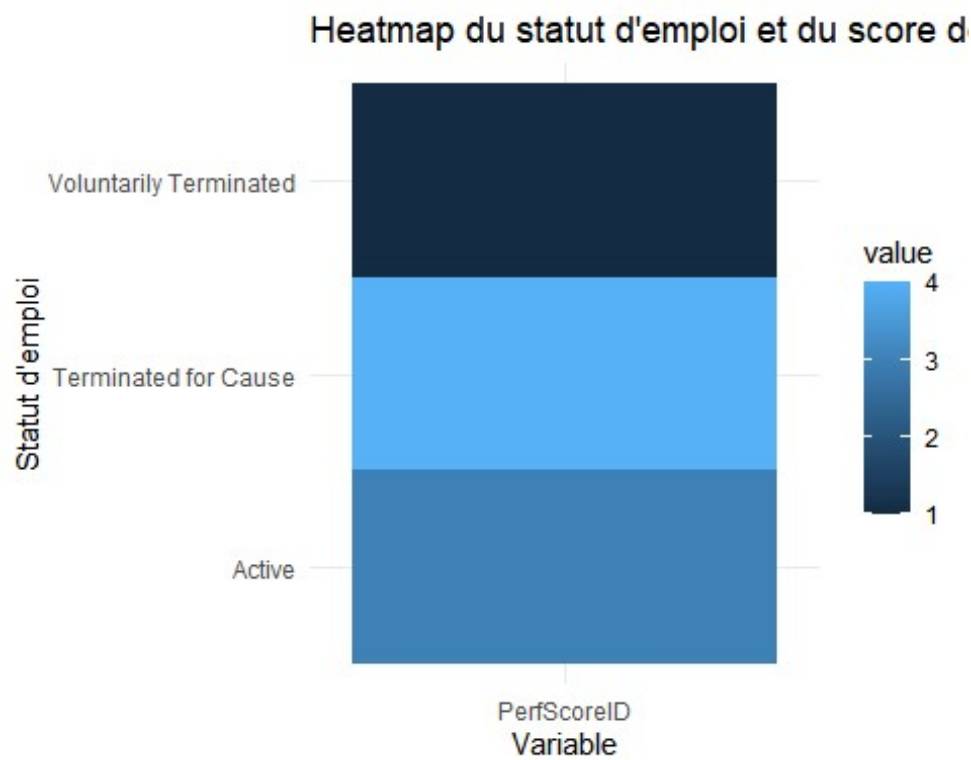
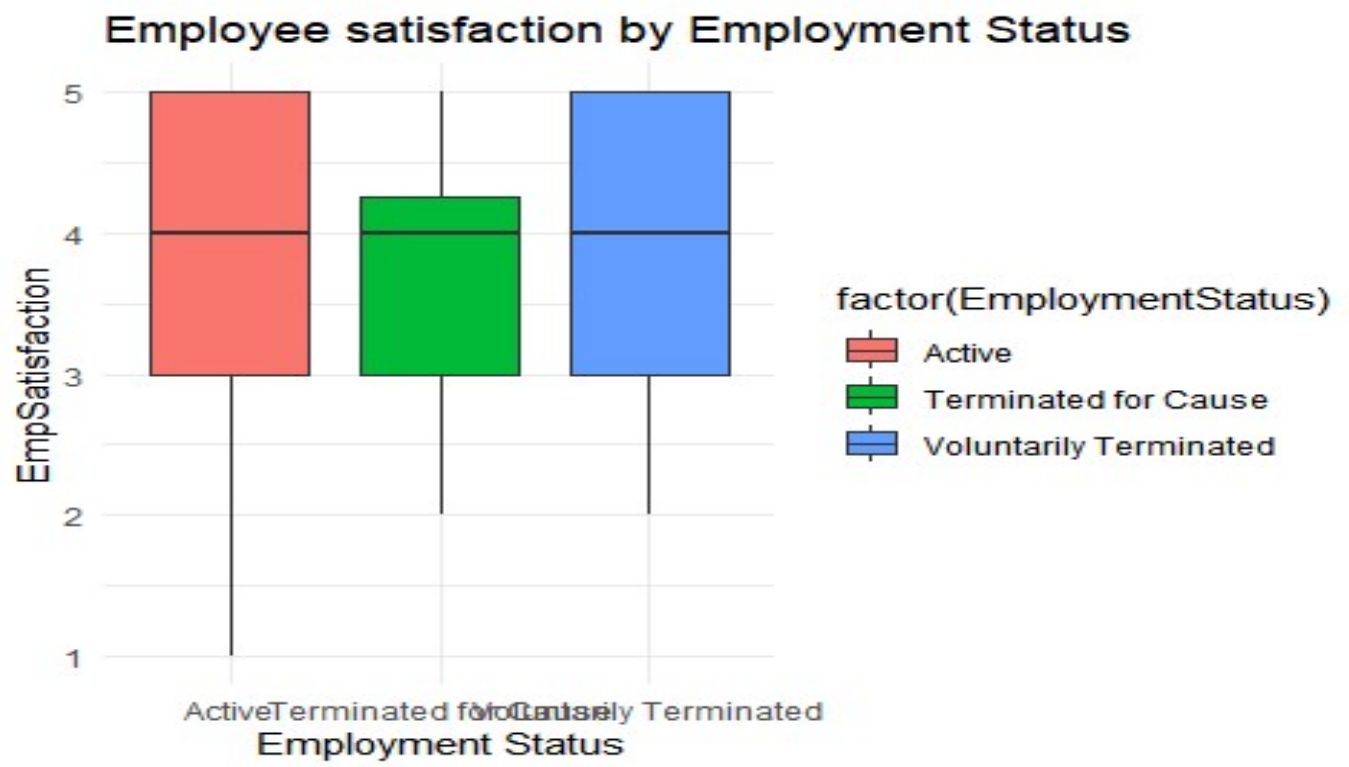


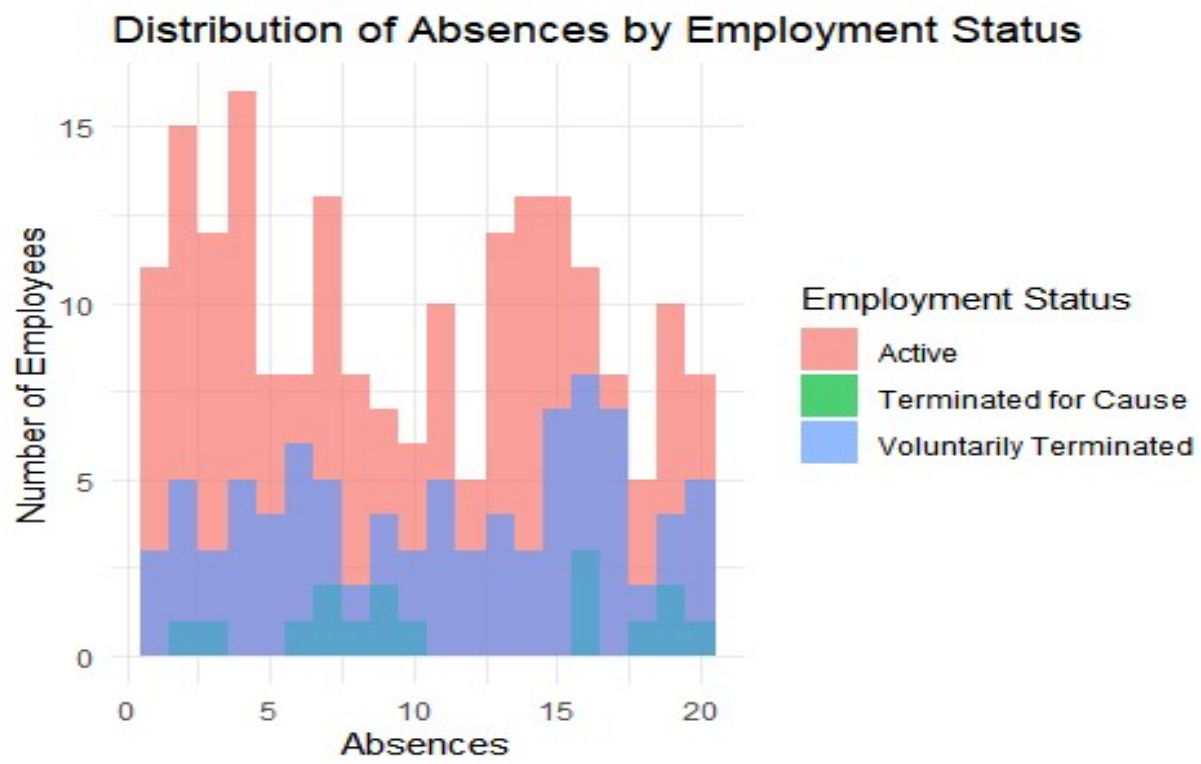
Salary Distribution by Employment Status



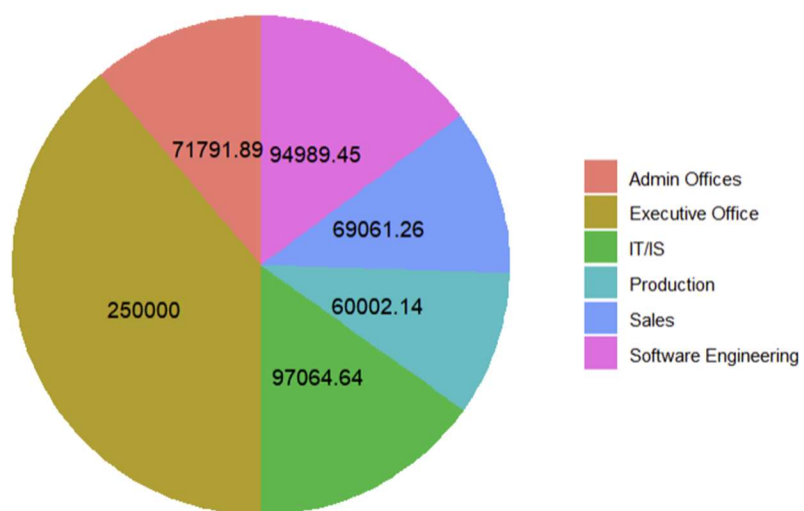
Employee Engagement by Employment Status







Average Salary by Department



Key Insights from Visual Data Analysis

A. Employee Performance by Manager

- The performance of employees significantly varies across different managers. Visual data reveals that some managers excel at nurturing high performers, while others struggle with a higher proportion of underperformers or those requiring performance improvement plans (PIPs).
- Specific findings include:
 - Manager 14 has a notably high number of employees who meet expectations fully, suggesting effective leadership or team dynamics.
 - Manager 12 stands out for having the highest number of employees exceeding expectations, indicating potentially exemplary leadership qualities or favorable work conditions.
 - Managers with a higher number of employees on PIPs may require additional training or support to enhance their team's performance.

B. Diversity Profile of the Organization

- The organization's workforce is predominantly White, with substantial underrepresentation of Hispanic, American Indian, and multiracial employees.
- The visual breakdown by race and ethnicity highlights critical areas for improvement in diversity and inclusion strategies.

C. Recruitment Sources and Diversity

- Analysis of recruitment sources shows that certain channels like LinkedIn and career fairs are more effective at attracting a diverse applicant pool.
- Recruitment efforts from online job applications and internal referrals show a higher proportion of White candidates, suggesting potential areas for diversifying recruitment strategies.

D. Predicting Employee Termination

- Visuals indicate that terminated employees, whether voluntary or involuntary, often have lower salaries compared to active employees, suggesting that salary may be a predictor of employee turnover.

- Engagement scores and employee satisfaction levels provide additional insights, with lower scores generally associated with terminated statuses. This correlation highlights the importance of maintaining high engagement and satisfaction to reduce turnover rates.
- Absence data further supports the likelihood of predicting terminations, particularly involuntary ones, where high absenteeism correlates with termination events.

E. Pay Equity Across Departments

- There are significant disparities in average salaries across different departments. The Executive Office notably has the highest average salary, far exceeding other departments like Production and Sales, which could raise concerns regarding pay equity.
- The visual representation of salary distributions underscores the need for a detailed review of compensation strategies to ensure fairness and equity.

Business Recommendations Based on Analytical Findings

1. Enhance Managerial Training:

- Focus on training managers with high numbers of underperformers or employees on PIPs to improve their leadership effectiveness.

2. Diversify Recruitment Channels:

- Expand the use of diverse recruitment sources that have proven effective in attracting a broad spectrum of candidates to improve overall workforce diversity.

3. Develop Retention Strategies:

- Implement targeted retention strategies focusing on competitive compensation, engaging work environments, and employee satisfaction to reduce turnover rates.

4. Review Compensation Policies:

- Conduct a thorough review of pay structures across departments to address and correct any inequities to ensure fair compensation practices throughout the organization.
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Conclusion

This detailed visual analysis of HR data provides valuable insights into how elements such as management effectiveness, diversity, recruitment sources, and pay equity impact the organization. By addressing the identified areas for improvement, the organization can enhance its HR strategies, fostering a more inclusive, equitable, and productive workplace. This report serves as a blueprint for strategic decision-making that can lead to significant improvements in organizational health and employee satisfaction.